

From Wins to Worth: A Quantitative Model of Performance, Attendance, Revenue, and Valuation for Sports Teams

Summary

This paper focuses on the transmission chain of "competitive performance - audience demand - revenue and profit - valuation - leverage decision" of professional sports clubs, and constructs a business decision-making model that can be updated in real time with season information, aiming to simultaneously increase team profits and value while controlling risks in an uncertain environment. We view the season as a "continuously updated system": every time a game is played, the strength, attendance, cash and leverage are updated, allowing the management to make decisions at any time during the season based on the same set of logic.

At the competitive level, this paper uses the Elo rating system to estimate the single-game winning rate and calculate the expected number of wins for the season; at the same time, it abstracts the injury of key players as a discrete downward shift of the team's effective strength, thereby obtaining the changes in winning scenarios under injury situations (the baseline expected number of wins is approximately 11.259, and the injury scenario drops to approximately 6.528).

At the commercial level, this paper establishes a seating demand prediction framework considering "seat occupancy truncation" and further proposes a segmented dynamic ticket pricing mechanism based on the predicted attendance rate, converting the seating prediction into box office revenue; under the baseline scenario, the estimated box office revenue is approximately 20.947528 (in million US dollars).

In terms of valuation, we calculated three sets of results simultaneously: revenue multiple, EBITDA multiple (closer to operating profit), and simplified DCF (cash flow discounting). The final valuation was obtained through weighted aggregation. Among them, the revenue multiple was only used for market benchmarking; what truly influenced the decision-making was the reflection of EBITDA/DCF on costs and interest rates, as well as the two hard constraints of the cash safety line and the leverage limit. Under the baseline scenario, the net profit is approximately 5.97m, the valuation is approximately 291.77m, and the end-of-period leverage is approximately 5.15%; an increase in interest rates will compress the valuation and passively push up leverage.

At the decision optimization level, this paper abstracts recruitment/transactions as discrete "actions", jointly optimizes actions and target leverage under constraints such as cash safety margin and debt limit, and performs segmented rolling re-optimization. The results show that the recommended strategy in the baseline scenario is NO_MOVE, with the target leverage approximately 0.05; the Monte Carlo risk assessment provides the percentile intervals of profit and valuation (for example, P10 of profit = 5.325160, P50 = 6.017653, P90 = 6.641407), and sensitivity analysis indicates that box office and attendance rate are the key factors simultaneously driving profit and valuation.

Key words: Elo Rating; Censored Demand Modeling; Dynamic Ticket Pricing; Team Valuation; Leverage Optimization

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I. Introduction

1.1 Background

Professional sports viewing is not merely a competitive activity; it is also an entertainment industry centered around the "attracting audiences - generating revenue - achieving profits" model. While team management strives for victory, they often have to prioritize commercial returns: Winning may bring more attention and revenue, but profits are not always solely determined by on-field performance. The popularity of players, marketing efforts, and the size of the market where the team operates will significantly affect ticket sales, peripheral products, and media revenue, etc. Therefore, to support teams in making executable business decisions during the season, it is necessary to connect "competitive data analysis" and "financial modeling" to form a dynamic decision-making system that can be updated over time.



Figure 1: Women's National Basketball Association



Figure 2: Indiana Fever logo

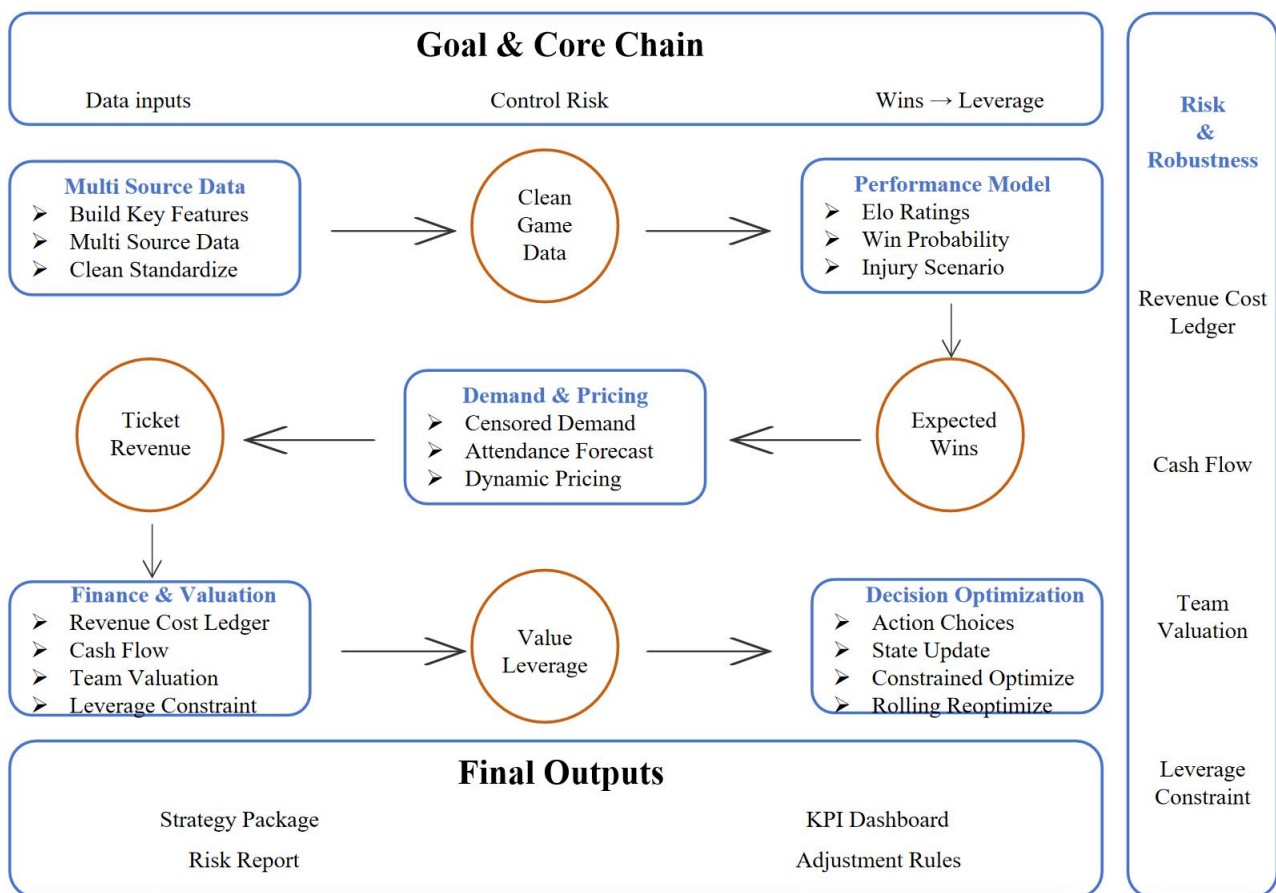
Under the league system, professional teams usually operate as part of the league's franchise, subject to various constraints such as salary caps, contract rules, scheduling and revenue distribution; at the same time, external factors such as interest rates, wage levels, injuries, and trading opportunities will constantly change during the season, making issues like "whether to use debt or equity financing" and "whether it is worth incurring higher costs to improve the record" highly dynamic and uncertain. They need to use systematic models to simultaneously evaluate competitive performance and financial results, thereby supporting the enhancement of team value and risk control.

1.2 Restatement of Problems

Based on the given requirements, we need to construct a business and management model for the selected professional team based on the publicly available competitive and financial data, and address the following key issues: First, establish a decision-making model that can be dynamically updated, enabling the management team to adjust the leverage level in response to changes in team performance and economic conditions, thereby maximizing profits and team value while controlling

risks; Second, propose the player acquisition strategy for the next season (such as draft, free market, trade, etc.) supported by the model, and discuss the advantages and disadvantages of this strategy for the commercial performance; Third, in the scenario of league expansion, analyze the impact of the characteristics of the new team's city and market on the original strategy and model output, and accordingly provide adjustment plans; Fourth, select and solve an additional operational decision problem (such as ticket prices, media, venues, or others), and explain how the model can assist the management team in responding and adjusting when key players suffer injuries.

1.3 Our works



II. Basic assumption

- **Assumption 1:** The team's business objective can be described as "maximizing profits and the value of the team (while controlling risks)".

Justification: We view the team as a subject that needs to make business decisions. The core goal of the management team is to increase profits and team value under constraints, while also paying attention to the risk level. Only in this way can the subsequent "optimal leverage and optimal strategy" have a clear optimization objective.

- **Assumption 2:** There exists a stable chain of transmission: winning rate/number of wins → attendance rate → revenue → profit/cash flow → valuation → leverage decision.
Justification: We assume that the performance in competitions will affect the demand (attendance), which in turn determines the box office revenue and total income. The income then flows into the financial module to generate profits and cash flow. Through the valuation module, the value of the team is obtained, which ultimately supports the dynamic adjustment of leverage and the output of decision-making.
- **Assumption 3:** The financial structure of the team can be approximated using a simplified ledger, and interest and debt have a linear relationship.
Justification: We approximate the profit as "EBITDA - interest", and assume that interest = debt × (market interest rate + credit spread), thereby enabling a direct connection between the operating results and the debt level, facilitating the assessment of optimal leverage and risk.
- **Assumption 4:** The leverage is defined by "debt/valuation" and is dynamically adjusted according to the rule of "target leverage - borrowing/repayment". At the same time, rigid constraints must be met.
Justification: We define leverage as debt/valuation. When the actual leverage deviates from the target, the team borrows or repays funds to bring it closer to the target, and must meet hard constraints such as the cash safety margin and leverage limit to ensure that the decisions output by the model are "executable" in reality.

III. Notations

Table 1: Notations

Symbols	Definition	Units
t	Competition/Time Number	None
p_t	Probability of Winning in the (t)th Game	None
W	Estimated Number of Wins for the Season (Sum of individual win probabilities)	Games
$\hat{A}_t$	Predicted Attendance for the (t)th Game	People
C	Capacity of the Venue (Upper Limit of Attendance)	People
$\bar{p}_t$	Average Ticket Price for the (t)th Game (Suggested value after dynamic pricing)	Dollars per ticket
R	Total Revenue (Box Office Revenue + Other Income)	Million Dollars (m)
Sal	Player Salaries	Million Dollars (m)
$EBITDA$	Operating Income (Used for profit calculation and valuation comparison)	Million Dollars (m)
D	Debt Balance	Million Dollars (m)
r	Loan Interest Rate (Can be understood as the market interest rate + spread)	1/year (or %/year)
Int	Interest Expense, commonly represented as $Int = D \cdot r$	Million Dollars (m)
Π	Profit (Simplified as $\Pi = EBITDA - Int$)	Million Dollars (m)
V	Team Valuation (Team Value)	Million Dollars

Symbols	Definition	Units
L	Leverage, ($L = D/V$)	(m) None

IV. Data Preprocessing

To ensure that the transmission chain from "winning probability or number of wins → attendance → revenue → profit and cash flow → valuation → leverage decision" can be stably supported by the same set of data, before entering the competitive prediction and demand prediction stages, this article first uniformly cleanses and aligns the competition-level data, ticketing attendance data, and venue information, so that the key variables are consistent in terms of scope, units, and symbol systems. In particular, the definition of the attendance prediction value $\hat{A}_i$ and the venue capacity C must strictly correspond to the same home venue and date scope of the same game.

Table 2: Data Source

Link	Data information
https://raw.githubusercontent.com/sportsdataverse/wehoop-data/main/wnba_schedule_master.csv	WNBA schedule (game dates, home and away teams)
https://api.seatgeek.com/2/events	SeatGeek ticket API (for extracting ticket prices)
https://herhoopstats.com/salary-cap-sheet/wnba/players/	WNBA players' salaries and salary caps
https://raw.githubusercontent.com/MJAlexander/nba/master/arenas_red.csv	NBA venue information (venues, cities, capacities)
https://fred.stlouisfed.org/graph/fredgraph.csv?id=DFE	Federal funds effective interest rate
https://www.basketball-reference.com/wnba/years/{year}_per_game.html	WNBA Player Season Statistics (by year; points scored, rebounds, assists, shooting percentage, etc.)
https://raw.githubusercontent.com/sportsdataverse/wehoop-data/main/wnba_stats_schedule_master.csv	WNBA Schedule + Game Index (used to correlate statistics with ticket prices)
https://raw.githubusercontent.com/sportsdataverse/wehoop-data/main/wnba/wnba_games_in_data_repo.csv	WNBA Game List (team names, team logos, etc.)
https://www.spotrac.com/wnba/cap/_/year/{year}	WNBA Team Salary Cap Ledger
https://fred.stlouisfed.org/graph/fredgraph.csv?id=BAMLH0A0HYM2	High-yield Bond Spread

4.1 Data Integration and Field Standardization

The integration of the dataset is conducted at the "single game" level as the minimum granularity. The schedule and score information are used as the main table, and the commercial side fields such as attendance numbers and venue capacity are merged under the same primary key. Considering that the public data may have slight discrepancies in date time zones and record time

points, this paper constructs the key value $k_i = (h_i, a_i, d_i)$ based on the home team, away team and game date, and allows for fault-tolerant matching within a ± 1 day range, thereby maximizing the available samples and reducing the situation where "the games should be played at the same venue but cannot be concatenated". This merging process can be expressed as a field mapping function for each game i :

$$(d_i, h_i, a_i) = \varphi(\text{raw_date}_i, \text{raw_home}_i, \text{raw_away}_i), k_i = (h_i, a_i, d_i), \quad (1)$$

Among them, $\varphi(\cdot)$ performs date parsing and normalization, team name text standardization, and the elimination of invalid team names and placeholder records, thereby ensuring that the "schedule table", "attendance table", and "venue table" can be stably linked under the same standard. At the same time, this paper follows a hierarchical organization method of "matches and schedules, requirements and ticketing, finance and macro" for the input, providing a consistent data entry point for subsequent modular modeling.

4.2 Missing Value Repair and Outlier Handling

The handling of missing values follows the principle of "first identifying the incorrect data format, then layer-by-layer repairing". Firstly, values that are obviously unreasonable are regarded as missing and uniformly set to null. For example, for any field x , using a rule

$$x \leq 0 \Rightarrow x \leftarrow NA \quad (2)$$

This way, the "0" is avoided from being mistakenly regarded as the actual number of seats or the actual capacity. Subsequently, a "same key priority, date fault tolerance as a fallback" repair strategy is adopted for the number of seats. The full-seat database is recorded as function $A^{all}(\cdot)$, and the original seat record in the main table is recorded as A_i^{orig} . Then, the repaired seats can be expressed as:

$$\tilde{A}_i = \begin{cases} A^{all}(d_i, h_i, a_i), \\ A^{all}(d_i + 1, h_i, a_i), \\ A^{all}(d_i - 1, h_i, a_i), \\ A_i^{orig}, \end{cases} \quad (3)$$

(Stop when the matching is completed from top to bottom according to this formula)

The capacity of the venue is repaired using a hierarchical median method of "venue priority, team backup, and overall conclusion". Let the venue be denoted as v_i , then:

$$\tilde{C}_i = \begin{cases} C_i, & C_i > 0, \\ \text{median}\{C_j: v_j = v_i, C_j > 0\}, & \text{If the set is non - empty,} \\ \text{median}\{C_j: h_j = h_i, C_j > 0\}, & \text{If the set is non - empty,} \\ \text{median}\{C_j: C_j > 0\}, & \text{Otherwise.} \end{cases} \quad (4)$$

In terms of handling outliers, this paper regards "rest days" as the most common and significant time-structured variable related to demand. Firstly, it calculates the difference in dates between two consecutive games within the same team's season. Then, extreme values are truncated to reduce noise propagation. Let $R_{i,t}$ represent the number of rest days for team t in game i .

$$R_{i,t} = \begin{cases} 10, & \text{If it is the team's first game of the season,} \\ \min(\max(d_i - d_{prev(i,t)}, 0), 10), & \text{Otherwise.} \end{cases} \quad (5)$$

After the above processing, the total sample size is 5065, and the capacity field is fully available. The attendance field still has a few missing values that are left blank to avoid introducing incorrect information. At the same time, based on the economic meaning that "being close to full capacity means the observation is truncated by the capacity limit", a "whether truncated by capacity" mark is retained for the subsequent demand model.

4.3 Feature Construction and Generation of Modeling Data Set

After the stitching and restoration are completed, this paper constructs a modeling data set using "a single game" as the observation unit, and explicitly defines the core variables required by the subsequent modules. Firstly, the full attendance rate $f_i = \tilde{A}_i / \tilde{C}_i$ is constructed to depict the demand intensity, and the truncated indicator variable is defined:

$$\delta_i = I(f_i \geq 0.97) \quad (6)$$

This ensures that when the observed seat occupancy approaches the capacity limit, this sample can be regarded as a "right-censored observation" in the demand module rather than a regular linear error.

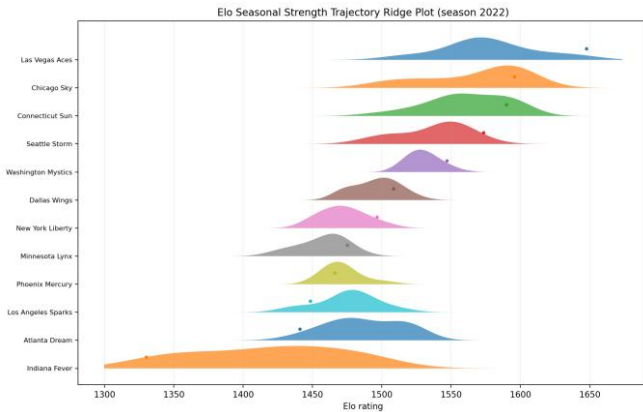


Figure 3: Elo Seasonal Strength Trajectory Ridge Plot

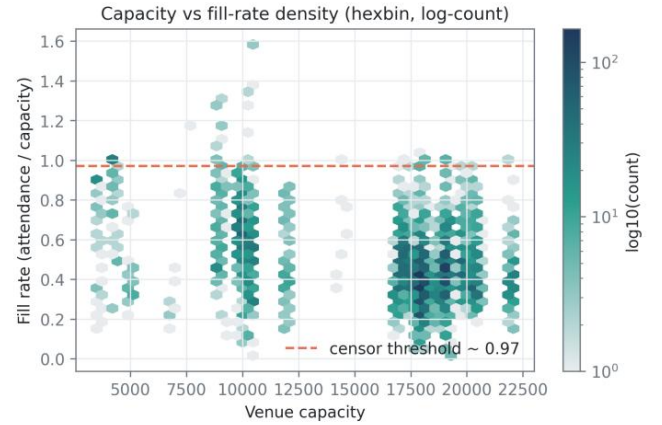


Figure 4: Capacity vs Fill-rate Density Plot

Secondly, generate the generation time and game schedule features, including month, week information, weekend indication, and the number of rest days for both home and away teams, etc., so that the feature vector of each game can be uniformly written as:

$$x_i = [1, month_i, weekend_i, R_{i,h_i}, R_{i,a_i}, opponent\ indicators, \dots]^T, \quad y_i = \tilde{A}_i, c_i = \tilde{C}_i \quad (7)$$

Finally, a training sample set for the demand module, denoted as $D = \{(x_i, y_i, c_i, \delta_i)\}$, is formed. A capacity limit is imposed on the seat prediction at the model output end to ensure consistency between the prediction and the definition.

$$\hat{A}_i^{final} = \min(\hat{A}_i, c_i), \quad (8)$$

Thus, it maintains a strict correspondence with the meanings of $\hat{A}_i$ and C in the symbol table, and is consistent with the module setting of "The demand forecast output is available and subject to capacity constraints".

V. Performance Prediction Model (Elo)

5.1 Elo Setup & Win Probability

After completing the data cleaning and generating the modeling dataset for the previous part, we regarded each game as an observation that occurred in chronological order, and set a dynamic strength score R for each team based on the game results, which updated as the game progressed. To ensure that different seasons are not disturbed by "historical residual states", we started each new season from the same baseline level, that is, the initial score of the season was set as $R = 1500$, and then it was updated in the course of the season based on the results of each game. For the i -th game, let the home team be h and the visiting team be a , and the pre-game scores be $R_{h,i}^e$ and $R_{a,i}^e$. According to the standard form of Elo, the probability of the home team winning is denoted as p_i , which is obtained by mapping the difference in scores between the two teams.

$$p_i = \frac{1}{1 + 10^{-(R_{h,i}^e - R_{a,i}^e)/400}} \quad (9)$$

Let the variable of the competition result be $y_i \in \{0,1\}$. In this case, if the home team wins, it is $y_i = 1$; if the home team loses, it is $y_i = 0$. The score update uses a fixed step size parameter K . In this paper, a set of settings with stable performance after backtesting is adopted, with $K = 20$. The post-match scores of the home and away teams are updated according to the following formula:

$$\begin{aligned} R_{h,i}^{st} &= R_{h,i}^e + K(y_i - p_i) \\ R_{a,i}^{st} &= R_{a,i}^e - K(y_i - p_i) \end{aligned} \quad (10)$$

We adopt rolling backtesting (using only past data to predict the future), and avoid using future information, thereby ensuring the reliability of the predictions. In this way, for each game, a key intermediate quantity p_i that can be directly used for the subsequent "winning rate/number of wins $\rightarrow$ attendance $\rightarrow$ revenue" is output, and it is consistent with the symbols in the table such as "single-game winning rate p_i " and "seasonal expected number of wins $\widehat{W}$ ".

5.2 Season Expected Wins and Injury Shock

Based on the single-game winning rate p_i , we define the expected number of wins for the season as the sum of the winning rates of all the games of the team during the season. Let the target team be T , and its season game set be $G(T)$. For any game i in the set, if T is the home team then take $\tilde{p}_{i,T} = p_i$, and if T is the away team then take $\tilde{p}_{i,T} = 1 - p_i$. So the expected number of wins for the season is:

$$\widehat{W}_T = \sum_{i \in G(T)} \tilde{p}_{i,T} \quad (11)$$

In our schedule prediction, we use the Elo score at the end of the previous season as the baseline strength input for the next season, thereby converting "team strength" into "winning rate per game" and further into "seasonal win expectancy". Taking the baseline scenario as an example, the model provides the target team's expected season win count of 11.259, corresponding to the baseline input in the subsequent revenue and valuation modules. Further, to depict the direct impact of core player injuries on the competitive aspect, we abstract injuries as a discrete reduction in the team's effective strength. That is, in the injury scenario, we apply a fixed negative displacement $\Delta_{inj} > 0$ to the team's Elo score, obtaining the effective score under injury conditions.

$$R_T^{inj} = R_T - \Delta_{inj} \quad (12)$$

And recalculate $\tilde{p}_{i,T}$ and $\widehat{W}_T$ for each game using R_T^{inj} . In the core injury scenario of this article, the injury caused the projected number of wins for the season to drop to 6.528, indicating that injuries will significantly alter the input conditions of "attendance and revenue" as well as "profit and optimal leverage decisions" in the future, thereby providing the management with quantifiable response basis.

5.3 Validation (simple metrics and calibration)

To verify the usability of Elo win rate prediction, we calculated three types of intuitive indicators on the historical game samples, evaluating the model output from three perspectives: probability error, probability logarithmic loss, and binary classification discrimination ability. The overall sample size was $n = 4810$, the Brier score was 0.234006, the LogLoss was 0.660180, and when using $p_i > 0.5$ as the "predict home team win" rule, the classification accuracy was 0.596881. At the same time, we further used the binning calibration method to test whether the predicted probabilities "match the reality", that is, grouping the predicted probabilities by intervals and comparing the average predicted probability with the actual win rate for each group. The results showed that in most intermediate probability intervals, the actual win rate was generally higher than the predicted value, demonstrating that the model's probability is conservative but still maintains a consistent directional nature.

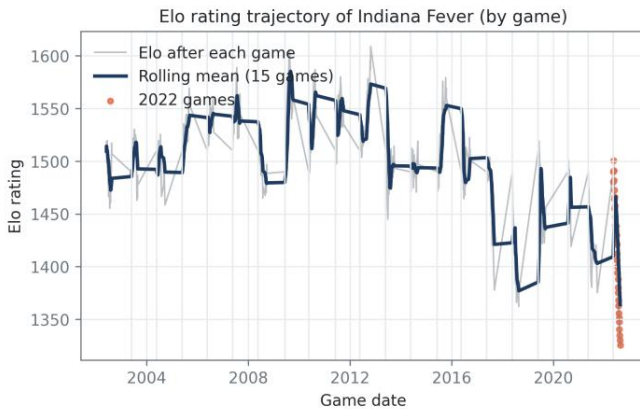


Figure 5: Elo Rating Trajectory of Indiana Fever

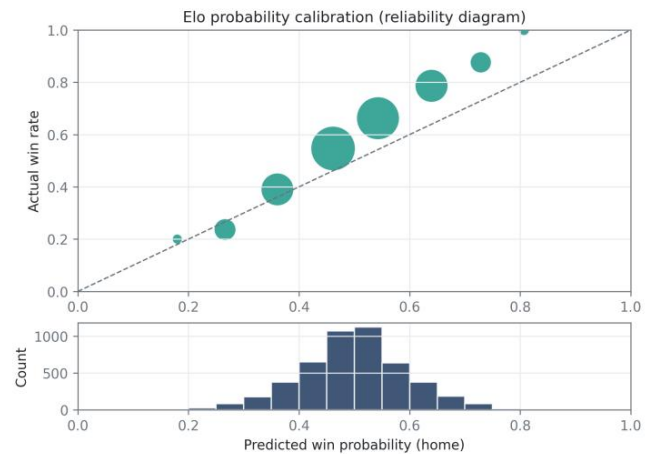


Figure 6: Elo probability Calibration

VI. Attendance and Ticket Pricing Model

6.1 Demand Model for Attendance

Following the transmission chain mentioned in the previous text, "winning rate or winning games will affect revenue through audience attendance", after we output the winning rate per game and the expected number of wins for the season at the competitive level, we use it as a key input at the audience end to depict real-world patterns such as "stronger matchups are more attractive", thereby forming a closed loop in the chain "winning rate or winning games $\rightarrow$ audience attendance rate $\rightarrow$ revenue $\rightarrow$ profit and cash flow $\rightarrow$ valuation $\rightarrow$ leverage decision" within the same model. To avoid information loss due to "full-seat games not revealing the true extent of demand", we record the actual audience demand for each game as a random variable D_t , while the observable attendance is recorded as A_t , and we introduce the upper limit of the venue capacity as C_t , using a simple but very realistic relationship to connect the two:

$$A_t = \min(D_t, C_t) \quad (13)$$

The occupancy rate model takes both the "strength end" (pre-match Elo winning rate, team/opponent rolling popularity) and the "time/schedule/surface end" (day of the week, month, whether it's a weekend, neutral venue, difference in rest days between the two teams) as exogenous features, and uses a chronological training-validation split (rolling window) to avoid using future information. Further assume that $\ln D_t$ follows a normal distribution, thus D_t follows a log-normal distribution, and use the eigenvector x_t to explain the systematic changes in the demand. Let's denote it as

$$\ln D_t \sim N(\mu_t, \sigma^2), \mu_t = x_t^\top \beta \quad (14)$$

When the competition is nearly full, the actual demand may have exceeded the capacity, but we can only know that "it is at least not less than the capacity", so it is regarded as a right-censored sample; when the competition is not nearly full, then we take $\ln A_t$ as the direct observation of $\ln D_t$. Based on this, we obtain β and σ using maximum likelihood estimation, and add ridge regularization to prevent the parameters from drifting in small samples or when there is collinearity. The overall log-likelihood is written as:

$$\begin{aligned} \ell(\beta, \sigma) = \sum_{t \in U} \left[-\frac{1}{2} \ln(2\pi) - \ln \sigma - \frac{(\ln A_t - \mu_t)^2}{2\sigma^2} \right] + \sum_{t \in C} \ln \left(1 - \Phi \left(\frac{\ln C_t - \mu_t}{\sigma} \right) \right) \\ - \lambda \sum_{j \geq 2} \beta_j^2 \end{aligned} \quad (15)$$

Here, U represents the non-deleted set, C represents the right-deleted set, and $\Phi(\cdot)$ is the standard normal distribution function. The regularization term does not penalize the intercept term to maintain scale interpretability. After obtaining the parameters, we do not directly use "average demand" as the upper seat prediction, but predict "expected upper seat considering the upper capacity limit", that is:

$$\hat{A}_t = E[\min(D_t, C_t) | x_t] \quad (16)$$

Its closed-form expression is:

$$\hat{A}_t = \exp\left(\mu_t + \frac{1}{2}\sigma^2\right) \Phi\left(\frac{\ln C_t - \mu_t - \sigma^2}{\sigma}\right) + C_t \left(1 - \Phi\left(\frac{\ln C_t - \mu_t}{\sigma}\right)\right) \quad (17)$$

In terms of feature construction, the model incorporates information such as the attractiveness of the opponent, the duration of the match, and the rolling popularity (James et al., 2021). It converts the output from the competitive end into a variable of "match suspense degree" and incorporates it into the demand end, enabling the audience prediction to naturally follow the previous winning rate module. In terms of verification results, the overall sample size is $n = 4617$. The MAE of the audience prediction is 1635.9533 people, the MAPE is 0.247843, and R^2 is 0.388249. Although the seating capacity prediction may have errors, we directly incorporated the residuals of this model into the subsequent Monte Carlo simulation, enabling the errors to be quantified clearly for box office, profit, valuation, and leverage using P10/P50/P90.

6.2 Dynamic Ticket Pricing Strategy (your extra decision)

After obtaining the predicted attendance figures for each game as $\hat{A}_t$, we regarded "dynamic pricing" as an executable additional operational decision within the season.

To be consistent with the symbol system, we denote $\hat{A}_t$ as the predicted attendance for the t -th game, C_t as the capacity, and $\hat{p}_t$ as the suggested average ticket price after dynamic pricing. First, we use the predicted full attendance rate to depict the "degree of supply and demand tension", and define:

$$\hat{\rho}_t = \frac{\hat{A}_t}{C_t} \quad (18)$$

Next, starting from the lowest ticket price p_t^{low} in the data, a stable "low ticket price to average ticket price" conversion coefficient is used to convert it into the benchmark average ticket price p_t^{base} , and a dynamic rate function $\varphi(\cdot)$ that varies with $\hat{\rho}_t$ is added on top of it, thereby obtaining the recommended average ticket price:

$$\bar{p}_t = p_t^{base} \cdot \varphi(\hat{\rho}_t), \quad p_t^{base} = m_0 p_t^{low} \quad (19)$$

The dynamic pricing rate is expressed using a segmented rule: raise the price when the venue is more likely to be fully occupied, and lower the price when it is less crowded to stimulate demand (Arslan et al., 2022). The ticket pricing decision incorporates price elasticity: the higher the ticket price, the lower the predicted attendance. Due to the lack of historical 'price - sales' data, we set the price elasticity as a set of scenarios, and expressed the demand as $\hat{A}(p; e)$ to explicitly describe the 'price increase will lower the attendance rate'. We calculate the optimal ticket price by maximizing the product of the ticket price and the predicted attendance (plus the revenue from season ticket conversions), rather than using a fixed segmented rate.

$$\varphi(\hat{\rho}_t) = \begin{cases} \alpha_{high}, & \hat{\rho}_t \geq 0.98 \\ \alpha_{mid}, & 0.90 \leq \hat{\rho}_t < 0.98 \\ \alpha_{low}, & \hat{\rho}_t \leq 0.65 \\ 1, & otherwise \end{cases} \quad (20)$$

Among them, $\alpha_{high} > \alpha_{mid} > 1$ is used for raising prices in cases of high demand, and $\alpha_{low} < 1$ is used for lowering prices in cases of low demand.

$$TicketRev = \sum_t \hat{A}_t \cdot \bar{p}_t, \text{ ticket_rev_m} = \frac{TicketRev}{10^6} \quad (21)$$

Under the baseline scenario, the model calculates the box office revenue as ticket_rev_m = 20.947528. At the same time, the increase in popularity and the impact of injuries will change $\hat{A}_t$, thereby altering $\hat{p}_t$ and $\bar{p}_t$, allowing the box office to exhibit visible fluctuations under different strategies and adverse scenarios.

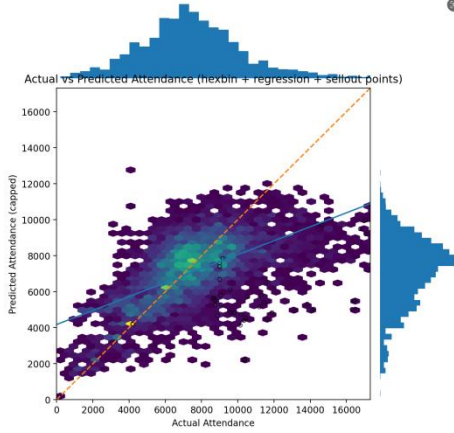


Figure 7: Actual vs Predicted Attendance Plot

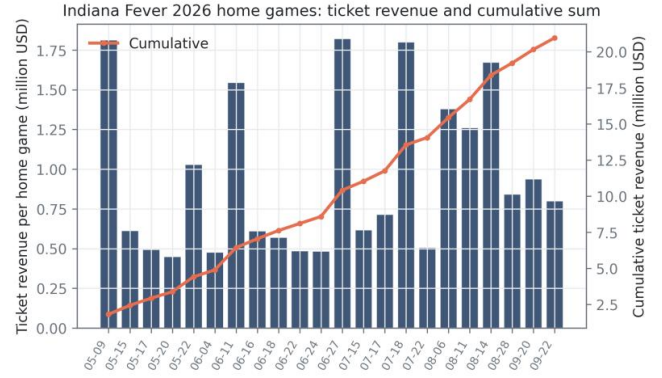


Figure 8: Indiana Fever 2026 Home Games

VII. Finance, Valuation, and Leverage

7.1 Profit/Cash Book (revenue–cost–interest)

Based on the fact that the model can already predict the number of wins and attendance rates game by game in the previous section, this part follows the stable transmission chain of "wins or win rate $\rightarrow$ attendance $\rightarrow$ revenue $\rightarrow$ profit and cash flow $\rightarrow$ valuation $\rightarrow$ leverage", strictly connecting the output of the competitive end to the financial end, thereby obtaining executable season profit and cash results. At the same time, interest and debt are linearly bound to directly link to the market interest rate scenarios (OpenStax, 2022). Specifically, let there be G games in the season, the predicted attendance for the t -th game be $\hat{A}_t$, and the suggested average price given by dynamic pricing be $\bar{p}_t$. Then, the box office revenue and total revenue can be written as:

$$T^{ticket} = 10^{-6} \sum_{t=1}^G \hat{A}_t \bar{p}_t, T = T^{ticket} + T^{other} \quad (22)$$

Here, T^{other} represents other revenues such as media and sponsorships, with the unit being millions of US dollars (m) (Deloitte, 2024). On the cost side, let the player salary be S^p , and the operating cost be approximated by the proportion of revenue ηT . Then, the operating profit indicator EBI can be written as :

$$EBI = T - (S^p + \eta T) \quad (23)$$

On the liability side, with the debt balance being D and the comprehensive interest rate being r , the interest expense satisfies a linear relationship:

$$I_D = D \cdot r \quad (24)$$

Thus, the profit is simplified to:

$$\pi = EBI - I_D \quad (25)$$

Cash flow follows the accounting closure relationship of "Ending balance = Beginning balance + Profit + Borrowing – Repayment", and a cash safety cushion constraint is "revenue - achieving profits" model (Fort, 2006). Under the base scenario, the model outputs the core financial results of the season, which include box office revenue of approximately 20.95m, other income of approximately 8.55m, total revenue of approximately 29.50m, operating profit EBI of approximately 7.38m, interest of approximately 1.41m, and net profit of approximately 5.97m, with the end-of-period cash maintained at the safety cushion level of 1.50m.

7.2 League Expansion and City Market Size (big/mid/small)

The valuation process is no longer dominated solely by revenue multiples: We use EBITDA multiples and simplified DCF as the main anchors (which can explicitly incorporate costs, interest rates and growth), while revenue multiples are only used for market benchmarking. The final valuation is obtained through a weighted average of the three factors and calibrated with scenario results to adjust the weights. Let the revenue multiple be m_{rev} , the EBITDA multiple be m_{ebi} , the interest rate discount function be $\varphi(r)$ and decrease as r increases. Then the revenue multiple valuation and EBITDA multiple valuation are respectively:

$$V_{rev} = m_{rev} \cdot T \cdot \varphi(r), V_{ebi} = m_{ebi} \cdot EBI \cdot \varphi(r) \quad (26)$$

Meanwhile, in order to provide a "directional verification" from the perspective of cash flow, we adopt the simplified DCF method. We discount the future free cash flows FCF_t over the next H years and add the terminal value term to obtain:

$$V_{DCF} = \sum_{t=1}^H \frac{FCF_t}{(1+r)^t} + \frac{FCF_H(1+g)}{(r-g)(1+r)^H} \quad (27)$$

Here, g represents the long-term growth rate. Under the baseline scenario, the valuations provided by the three methods are approximately 291.77m for the revenue multiple, 88.50m for the EBITDA multiple, and 55.87m for the simplified DCF. Although the absolute numerical values are different, they can be used to verify the consistency of the direction of the same strategy under different valuation metrics, thereby supporting the statement "the conclusion is robust". Further, in the aggressive heavy transaction scenario, due to the increase in revenue, the revenue multiple valuation rises to approximately 338.46m. However, due to the EBIT turning negative, the EBITDA multiple valuation and DCF valuation respectively become approximately -5.33m and -33.12m. This indicates that simply pursuing "high valuations in the revenue metric" will be significantly punished by the "profit and cash flow metrics", thereby naturally forming an explanation for the risk constraints.

It particularly highlights the significant differences in "major transactions" strategies between the revenue calculation basis and the cash flow calculation basis, thereby providing intuitive evidence for the subsequent "balancing profit and value under constraints".

7.3 Key Player Injury Response (how decision changes)

After obtaining the valuation V , we express the leverage as the ratio of debt to valuation, and

implement dynamic adjustment using the "target leverage-driven borrowing and repayment" rule. At the same time, we ensure the strategy's feasibility through hard constraints such as the cash safety cushion. Let the actual leverage be L and the target leverage be L^* , then:

$$L = \frac{D}{V}, D^* = L^*V \quad (28)$$

If the current debt level is low, then borrow to make up the shortfall. If it is high, repay within the limit of not exceeding the cash safety margin, so that the end-of-period debt converges towards D^* . The closed-updating formula can be written as:

$$D_{end} = D + \max(0, D^* - D) - \min(\max(0, D - D^*), Cash_{begin} + \pi - Cash_{min}) \quad (29)$$

And it also meets $Cash_{end} \geq Cash_{min}$. Under the baseline scenario, the model provides a target leverage of approximately 5.12%, the end-of-period debt is approximately 15.03m, the achieved end-of-period leverage is approximately 5.15%, and the end-of-period cash is maintained at a safety cushion level of 1.50m. Under external interest rate shocks, this rule will automatically transmit "changes in the financing environment" to profits and valuations, and then feed back to the leverage results. For example, when the interest rate rises by one percentage point, the net profit drops to approximately 5.74m, and the valuation drops to approximately 262.58m. Thus, under similar debt levels, the end-of-period leverage rises to approximately 5.81%. When the interest rate drops by one percentage point, the net profit rises to approximately 6.19m, and the valuation rises to approximately 333.54m, allowing the end-of-period debt to rise to approximately 23.74m and the leverage level to be approximately 7.12%. Similarly, when core injuries lead to a decrease in expected wins and further depress total revenue and valuation, the model outputs a net profit of approximately 4.70m, a valuation of approximately 241.72m, and an end-of-period leverage of approximately 6.74%. This shows that even in unfavorable competitive scenarios, leverage will rebalance with changes in valuation and maintain the feasibility of decision-making through the cash safety cushion constraint. Executable trigger rule: If the predicted occupancy rate drops by $\geq 15\%$ compared to the benchmark or the projected end-of-period cash approaches the safety margin, immediately reduce the target leverage and suspend high-cost recruitment until the key indicators recover.

VIII. Decision Optimization and Player Acquisition

8.1 Player Actions (no move / sign / trade) as "buttons"

In order to convert "signing or trading" into calculable management decisions, we abstract all possible player acquisition schemes into a set of actions A . For any action $a \in A$, its direct impact is characterized by a triplet: additional salary expenditure $\Delta C(a)$, increase in competitive intensity $\Delta E(a)$, and increase in popularity $\Delta P(a)$. At the decision point t , the team's state is denoted as $s_t = (E_t, P_t, cash_t, D_t)$, where E_t represents Elo strength, P_t represents popularity level, $cash_t$ and D_t are cash and debt respectively. After executing the action, the state is updated to:

$$E_t^{(a)} = E_t + \Delta E(a), P_t^{(a)} = P_t + \Delta P(a), C_t^{(a)} = C_t + \Delta C(a) \quad (30)$$

Subsequently, using the pre-determined win rate prediction and demand income modules, $(E_t^{(a)}, P_t^{(a)})$ is mapped to season win rate $w_{rt}(a)$, income $R_t(a)$, profit $\Pi_t(a)$ and valuation $V_t(a)$, thereby enabling "actions" to affect cash, valuation and leverage levels through the same pipeline.

8.2 One-time constraint optimization

Given the initial information at the beginning of the season, we combine the action selection and leverage target into a constrained optimization problem. Let ℓ be the target leverage. Define the end-of-period debt $D_{t,end}(a, \ell)$ and the end-of-period cash $cash_{t,end}(a, \ell)$ as generated by the financial rolling equation, and characterize the risk using quantiles. We use the median of the worst quantile's drawdown as the risk measure:

$$\Delta_{risk}^{\Pi} = Q_{0.50}(\Pi_t) - Q_{0.10}(\Pi_t), \Delta_{risk}^V = Q_{0.50}(V_t) \quad (31)$$

The final joint decision is given by the following formula, where α , λ , and β control the trade-off between "returns and robustness":

$$(a_t^*, \ell_t^*) = \operatorname{argmax}_{a \in A, \ell \in [\ell_{min}, \ell_{max}]} E[V_t(a, \ell)] + \alpha E[\Pi_t(a, \ell)] - \lambda(\Delta_{risk}^{\Pi} + \beta \Delta_{risk}^V) \quad (32)$$

$$s.t. \text{ cash}_{t,end}(a, \ell) \geq \text{cash}_{min}, \quad 0 \leq D_{t,end}(a, \ell) \leq D_{max}, \quad (33)$$

$$\frac{D_{t,end}(a, \ell)}{V_t(a, \ell)} = \ell. \quad (34)$$

Under the baseline scenario, the one-time optimal output is the optimal action $a_t^* = NO_MOVE$, the optimal target leverage $\ell_t^* = 0.05$, and the winning rate $w_{r_{used}} = 0.255886$ is used accordingly. The Monte Carlo distribution shows that the profit quantiles are $P10 = 5.325160, P50 = 6.017653, P90 = 6.641407$, and the valuation quantiles are $P10 = 266.386541, P50 = 293.781896, P90 = 318.457876$. The profit drawdown indicator $P50 - P10 = 0.692494$ indicates that there is still a positive profit even at the worst quantile and the risk is quantifiable.

8.3 Rolling optimization

The season information will be gradually revealed. To avoid "reviewing the answers afterwards", for each prediction and re-optimization, only the ongoing matches and known information at that time are used, and no future match data is used in advance. We divide the season into four segments and update the status and decisions in a rolling manner. Let the initial state of the k-th segment be $s_{t,k}$. After the results of the matches within the segment are updated with Elo and demand-side information, the state is recursively transformed into:

$$s_{t,k+1} = T(s_{t,k}, a_{t,k}^*, \ell_{t,k}^*; Data_{t,k}) \quad (35)$$

The operator $T(\cdot)$ is defined jointly by the win rate prediction, the demand income model, and the financial ledger, and is re-solved for each section to obtain $(a_{t,k}^*, \ell_{t,k}^*)$, with the same type of

constraints optimization as in the previous section. The rolling results of the benchmark scenario show that the four optimal decisions remain as NO_MOVE and $\ell = 0.05$. Meanwhile, due to the accumulation of profits into cash, the median of cash will increase continuously as the sections progress, demonstrating the robust characteristic of "information update but strategy stability".

8.4 Recommended benchmark strategy and key indicators

Under the baseline scenario, by combining one-time and rolling optimizations, it is recommended to adopt the default strategy: NO_MOVE, with a target leverage of approximately 5%. This approach aims to achieve a balanced consideration of profits and valuation within the constraints of the cash safety cushion and debt limit. To prevent "always remaining unchanged", this study provides executable trigger-response rules:

(i) Cash warning: If $Cash_{end} \leq 1.2 \times Cash_{min}$, reduce the target leverage to 3%–4% and suspend new debt issuance;

(ii) Demand warning: If the predicted attendance rate for two consecutive segments is lower than the benchmark (or below the preset threshold), prioritize the "popularity reinforcement/marketing synergy" strategy with SIGN (high ΔP , low ΔS) to restore the revenue side;

(iii) Competition warning: If injuries or performance issues lead to a significant downward revision in the predicted winning games (below the target range), prioritize the "immediate strength reinforcement" strategy with SIGN/TRADE (high ΔE), while simultaneously tightening the leverage to control the downside risk.

These rules ensure: no random movements during normal times, action when triggered, and each action can be quantitatively explained in the model. The injury situation is modeled as a probabilistic process that can be updated continuously (each injury probability p , and the distribution of absence games after injury L). The latest attendance/injury data is used to update it at each stage. Only when the trigger threshold is reached will it switch to SIGN/TRADE.

The rolling review only requires monitoring four core indicators: competitive side (E_t , predicted win rate/success rate, injury impact), demand side (predicted attendance and suggested ticket price, ticket revenue), financial side (profit, $Cash_{end}$, $Debt_{end}$, actual leverage ℓ), and risk side (P50 - P10). When the indicators reach the warning threshold of 8.4.1, the corresponding actions are initiated; otherwise, the default strategy is maintained to minimize unnecessary salary and leverage fluctuations.

To make the recruitment process calculable and comparable, any candidate player (or transaction package) will be uniformly represented as (ΔS , ΔE , ΔP), and ranked on the same scale using a concise scoring system:

$$Score = \omega_E \Delta E + \omega_P \Delta P - \omega_S \Delta S \quad (36)$$

We regard SIGN as free agent signings, TRADE as player transactions, and DRAFT as draft selections; and we strictly enforce the salary cap, the maximum roster limit, and the cash safety line. Any action that fails to meet these requirements is directly ruled as unfeasible.

Strategy suggestions:

1) Drafting: Prioritize selecting players with high ΔE and low ΔS (i.e., combat

power/bargainability) to stabilize the winning rate and retain cash reserves;

2) Free market: If demand is weak, prioritize high ΔP (to drive attendance and revenue); if the competition is weak, prioritize high ΔE (to directly boost the winning rate), but it must meet the cash safety margin;

3) Trading: Only activate when "both competition and demand deteriorate simultaneously", prioritize transactions with controllable ΔS , and simultaneously lower the target leverage to resist passive leverage increase caused by downward valuation.

At each stage, we present a small comparison table: three rows labeled as NO_MOVE, SIGN, and TRADE. We use the same set of constraints and objective functions to score them, and clearly state why this particular row was chosen in the end.

Table 3: Decision Comparison Table

Indicators	[NO_MOVE]	[SIGN] SIGN_STAR	[TRADE] TRADE_SUPERSTAR
Expected wins (games)	11.26	14.86 (+3.61)	16.28 (+5.02)
Average attendance rate (estimated)	45.15%	49.67% (+4.52pct)	52.38% (+7.22pct)
Total home audience (estimated, people)	171,586	188,745 (+17,159)	199,040 (+27,454)
Box office revenue (millions)	20.95	25.35 (+4.40)	28.19 (+7.24)
Other income (millions)	8.55	8.92 (+0.36)	9.12 (+0.56)
Total revenue (millions)	29.5	34.27 (+4.76)	37.30 (+7.80)
Extra salary/investment (millions)	0	6	9
Net profit (millions)	5.97	1.16 (-4.81)	-1.08 (-7.05)
Valuation (millions)	291.77	338.89 (+47.12)	368.95 (+77.18)

To make the actions "visible", we interpret SIGN_STAR as "signing an all-star level free agent", with the increment being $\Delta S = +3.61$ wins, $\Delta E = +4.40m$, and $\Delta P = +4.40m$; and we interpret TRADE_SUPERSTAR as "replacing an MVP-level player with a rotation player and a draft pick", with the increment being $\Delta S = +5.02$, $\Delta E = +6.10m$, and $\Delta P = +0.541m$. The incremental parameters of the players' actions (salary increment, Elo_boost, pop_boost) can be set in a repeatable manner according to "range intervals", and sensitivity tests are conducted on ranges such as Elo_boost = +50 to +100 to ensure that the conclusion is not dependent on a single subjective point.

IX. Scenario Analysis

For any scenario s , first, the winning probability of the competition is used to obtain the expected number of wins for the season, then the number of wins is connected to the demand-side model to the revenue side, subsequently entering the financial ledger and valuation modules, and finally using the "target leverage-driven borrowing and repayment" rule to feedback external shocks to the end-of-period debt and leverage. Formally, the scenario parameters are written as $\theta^{(s)} = (r^{(s)}, \Delta S_{inj}^{(s)}, \gamma_{market}^{(s)})$, and the number of wins and revenue can be written as

$$\widehat{W}^{(s)} = \sum_{i \in G} \hat{p}_i^{(s)}, \hat{p}_i^{(s)} = \sigma(\text{EloDiff}_i^{(s)}) \quad (37)$$

Among them, injuries are incorporated into the winning rate calculation through a discrete

downward shift of the effective intensity.

$$S_{eff}^{(s)} = S - \Delta S_{inj}^{(s)} \quad (38)$$

In the financial aspect, let the total revenue be $R^{(s)}$, the player salary be $S_p^{(s)}$, the operating cost be approximately $BR^{(s)}$ based on the revenue ratio, the debt balance be $D^{(s)}$, and the comprehensive interest rate be $r^{(s)}$. Then, the operating profit, interest, and net profit satisfy:

$$EBI^{(s)} = R^{(s)} - S_p^{(s)} - BR^{(s)}, I^{(s)} = r^{(s)} D^{(s)}, \pi^{(s)} = EBI^{(s)} - I^{(s)} \quad (39)$$

Interest rate changes enter the valuation through two paths: on the one hand, they alter the discount rate for DCF and simultaneously affect the interest expense on debt; on the other hand, they pass on the cost pressure to the valuation through EBITDA and cash flow. Therefore, even if the revenue multiple remains unchanged, an upward interest rate will lower the valuation and passively increase leverage, thereby triggering a more conservative target leverage setting.

$$V^{(s)} = m_{rev} \cdot R^{(s)} \cdot \phi(r^{(s)}) \quad (40)$$

Thus, the leverage is defined as $\ell^{(s)} = D^{(s)}/V^{(s)}$, and the target leverage ℓ^* is used to determine borrowing and repayment, so that the end-of-period debt converges to $\ell^* V^{(s)}$ without exceeding the cash safety margin constraint.

9.1 Interest Rate Shock (up/down)

The interest rate shock enters both the profit side and the valuation side in the model. The mechanism is "linear increase in interest rates and decrease in discount factors", which then pushes up leverage when the debt level approaches and changes the optimal debt final value when the financing rules permit. Based on the baseline scenario, the model provides a baseline net profit of approximately 5.97m, valuation of approximately 291.77m, end debt of approximately 15.03m, and end leverage of approximately 5.15%. When the interest rate rises by 1%, the net profit drops to approximately 5.74m, the valuation drops to approximately 262.58m, the end debt is approximately 15.26m, and thus the leverage is passively raised to approximately 5.81%, demonstrating the amplification effect of "valuation compression leading to a smaller denominator". When the interest rate drops by 1%, the net profit rises to approximately 6.19m, the valuation rises to approximately 333.54m, and at the same time, due to the financing rules driven by the target leverage, the financing rules are more "willing to increase leverage", the end debt rises to approximately 23.74m, the end leverage reaches approximately 7.12%, and a higher end cash balance is formed.

9.2 League Expansion and City Market Size (big/mid/small)

We selected 5 candidate cities for league expansion and ranked them as follows: Toronto, Philadelphia, Portland, Nashville, and Louisville. We scored them using four indicators (market size, travel cost, competition intensity, sponsorship/media space) and mapped the scores to three market categories: BIG, MID, and SMALL. The model showed that BIG was the best (with an estimated value of approximately 300.23m) > MID (about 291.77m) > SMALL (about 285.01m). The

expansion of the league mainly affects the expected number of wins by altering the schedule and the intensity of competition. This, in turn, is transmitted to profits and valuations through the demand and revenue modules. The size of the urban market is entered into the revenue side through the exogenous scaling of "non-theatrical revenue" (with the scaling coefficient of $other_rev$ denoted as $(1 + \gamma_{market})$), then

$$R^{(s)} = ticket_rev + (1 + \gamma_{market}^{(s)}) \cdot other_rev \quad (41)$$

In the BIG (+10%) market, the total revenue is approximately 30.36m, the net profit is approximately 6.18m, the valuation is approximately 300.23m, and the end debt is approximately 15.37m. There is additional accumulation above the cash safety cushion. In the MID (0%) market, it returns to the benchmark. In the SMALL (-8%) market, the total revenue is approximately 28.82m, the net profit is approximately 5.80m, the valuation is approximately 285.01m, and the end debt is approximately 15.20m. The leverage is more likely to increase relatively, thus indicating that "small markets require more conservative financing and stronger cash discipline". For the expansion plan, we have identified 3 to 5 candidate cities.

9.3 Key Player Injury Response (how decision changes)

The injuries of key players are abstracted in the model as a discrete downward shift in effective intensity, systematically reducing the winning rate and the expected number of wins for the season, and thereby compressing the attendance and revenue. In the scenario of core player injuries, the expected number of wins drops to 6.528, the total revenue is approximately 24.44m, the net profit is approximately 4.70m, the valuation is approximately 241.72m, but the end debt is approximately 16.30m, causing the end leverage to rise to approximately 6.74%. This indicates that injuries will passively amplify financial risks through "a decrease in the denominator of the valuation". Therefore, changes in management decisions should not be to pursue more aggressive high-salary transactions, but to actively lower the target leverage within the existing leverage loop and prioritize maintaining the cash safety cushion, so that the debt converges to a lower ℓ^*V level, thereby bringing the passively rising leverage back within a controllable range. This is consistent with the previous conclusion that the risk of aggressive heavy transactions "seems to have a higher valuation but has negative profits and higher debt" is that injuries will further exacerbate the penalty of such strategies in terms of profit and cash flow indicators.

X. Uncertainty, Monte Carlo, and Sensitivity

10.1 Monte Carlo Risk: P10 / P50 / P90 (profit, valuation)

The Monte Carlo part takes the optimal strategy as the fixed decision input, that is, the action selection is NO_MOVE and the target leverage is $\ell^* = 0.05$. On this basis, random sampling is conducted on the key disturbances at the revenue and cost ends to obtain the empirical distribution of profits and valuations. Let the random disturbance vector of a single simulation be $\varepsilon^{(k)}$, and apply it

to box office and other revenues as well as costs or injury-related shocks. Then, the total revenue and profit valuation of the k th sampling can be written as:

$$R^{(k)} = \text{ticket_rev}(\varepsilon^{(k)}) + \text{other_rev}(\varepsilon^{(k)}) \quad (42)$$

$$\pi^{(k)} = (R^{(k)} - S_p - BR^{(k)}) - rD \quad (43)$$

$$V^{(k)} = m_{rev} \cdot R^{(k)} \cdot \phi(r) \quad (44)$$

After obtaining $\{\pi^{(k)}\}_{k=1}^K$ and $\{V^{(k)}\}_{k=1}^K$, the downside risk and median return are characterized using empirical quantiles. Their mathematical definitions can be written as

$$P_q(X) = \inf \left\{ x: \frac{1}{K} \sum_{k=1}^K 1(X^{(k)} \leq x) \geq q \right\} \quad (45)$$

Based on the code output, the profit distribution under this optimal strategy has $P10 = 5.325160$, $P50 = 6.017653$, and $P90 = 6.641407$. The valuation distribution has $P10 = 266.386541$, $P50 = 293.781896$, and $P90 = 318.457876$. The risk span indicator $P50 - P10 = 0.692494$ indicates that there are identifiable downward fluctuations in the distribution rather than being wrongly flattened to a constant. The Monte Carlo simulation results show that even in the worst-case scenario of P10, the profit remains positive and the cash does not fall below the safety margin. Therefore, this strategy can still be executed under risk.

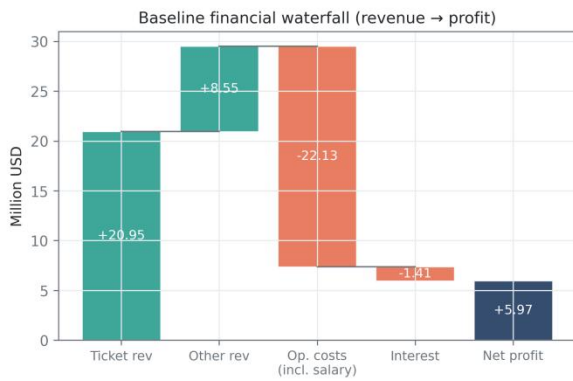


Figure 9: Baseline Financial Waterfall Plot

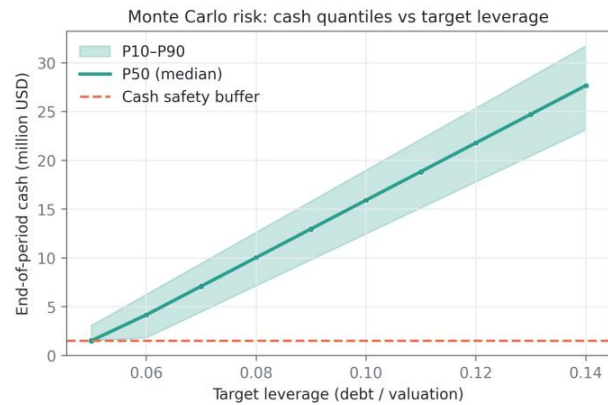


Figure 10: Monte Carlo risk Plot

10.2 Sensitivity Analysis (ticket, cost, media, injury, etc.)

The sensitivity analysis uses the single-factor perturbation method to make symmetrical or scenario-based changes to the key assumptions and map the resulting output variations back to the profit and valuation ends to identify the most critical drivers that need to be managed. Taking the baseline output as (π_0, V_0) , the output after applying a relative perturbation δ to any input factor x is $(\pi(x(1 + \delta)), V(x(1 + \delta)))$. The sensitivity results are as follows:

$$\Delta\pi = \pi(x(1 + \delta)) - \pi_0, \Delta V = V(x(1 + \delta)) - V_0 \quad (46)$$

To present and determine the priority of the code results based on the magnitude of the variation, the results show that the impact on the box office is the most direct and has a significant effect on the

valuation. When the box office drops by 20%, the profit decreases by 1.047376 and the valuation decreases by 41.434670. When the box office increases by 20%, the profit increases by 1.047376 and the valuation increases by 41.434670. Similarly, injuries and the decline in attendance have a strong impact. When attendance drops by 18%, the profit decreases by 1.049564 and the valuation decreases by 41.521203. When attendance drops by 12%, the profit decreases by 0.735351 and the valuation decreases by 29.090802. The $\pm 20\%$ perturbation of media and sponsorship and other_rev has an impact on the profit and valuation of ± 0.427700 and ± 16.920000 respectively, while the $\pm 20\%$ perturbation of the cost ratio has an impact on the profit of ± 4.425229 . However, under the valuation method based on "income multiple", the valuation change is zero. Because "income multiple valuation" is inherently not sensitive to cost impacts, we only use it as a reference for market benchmarking; the actual hard constraints used for decision-making are the "realistic test" composed of EBITDA/Simplified DCF and the cash safety line, as well as the leverage ceiling. Any plan that results in negative EBIT or touches the cash bottom line is directly ruled out as unfeasible. This indicates that costs mainly affect the feasibility through profit and cash constraints, and the valuation's reflection of costs requires relying on EBITDA multiples or simplified DCF and other comparison metrics. Thus, it is also consistent with the previous writing of "cross-validation of multiple valuation methods". The sensitivity analysis provides clear priorities: focus first on the attendance rate / box office (which affects both profit and valuation), then on the cost rate (which determines the cash risk line), and other factors are dealt with after setting the trigger thresholds.

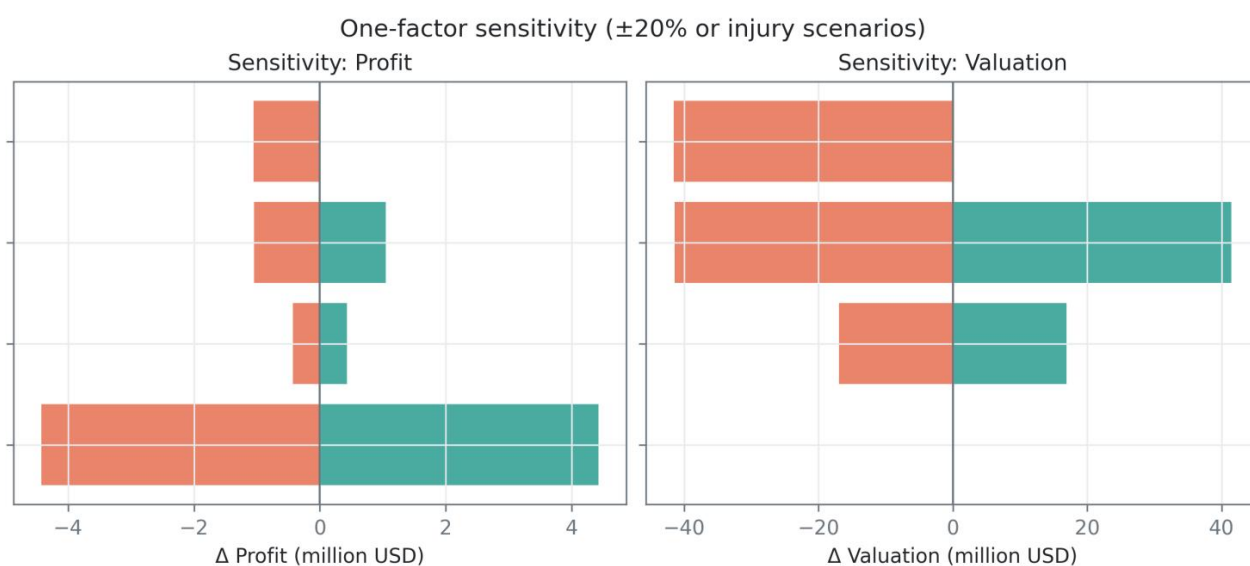


Figure 11: One-factor Sensitivity Plot

The cost ratio has the strongest impact on profit, but it is not sensitive to the main valuation method. Thus, it provides a clear priority conclusion for management: "First manage the risks of box office and attendance, then manage cost discipline, and at the same time, check the indirect impact of costs on value using a comparison valuation method".

Table 4: Risk Comparison Table

Risk indicators (the more conservative, the better)	[NO_MOVE]	[SIGN] SIGN_STAR	[TRADE] TRADE_SUPERSTAR
Corresponding leverage level (lev_target)	0.05	0.07	0.08
Profit P10 / P50 / P90 (in	5.33 / 6.02 / 6.64	0.21 / 1.03 / 1.75	-2.476813966

Risk indicators (the more conservative, the better)	[NO_MOVE]	[SIGN] SIGN_STAR	[TRADE] TRADE_SUPERSTAR
millions)			
Cash P10 / P50 / P90 (in millions)	1.50 / 1.50 / 3.06	1.79 / 4.90 / 7.62	4.00 / 7.58 / 10.75
Valuation P10 / P50 / P90 (in millions)	266.39 / 293.78 / 318.46	301.23 / 333.80 / 362.40	322.10 / 356.07 / 386.22
Downward risk range (P50 - P10, the larger, the more unstable)	0.69	0.82	0.86

XI. Model Evaluation and Conclusion

11.1 Strengths

- This model forms a complete closed loop: it first uses win rate prediction and attendance rate prediction to drive revenue, then calculates profits and cash flow using the financial ledger, and finally makes decisions under valuation and leverage constraints. Thus, it can clearly explain the cause-and-effect relationship and can be directly implemented.
- The predictions of this model are testable: win rate is measured for reliability using indicators such as accuracy rate, and attendance rate is measured for explanatory power using error and fit degree. Therefore, the conclusion is not "made on a whim".
- The decisions of this model are consistent and executable: the cash safety margin and leverage limit are written into hard constraints, so the recommended strategy naturally meets the implementation conditions.
- The robustness of this model is traceable: through rolling updates and scenario verification, it can explain how external shocks affect profits, valuations, and leverage step by step.
- This model does not rely on a single indicator for valuation: income multiples and cash flow-based valuations are compared simultaneously, thereby reducing the risk of misjudging as "appearing good on the surface but not sustainable".

11.2 Weaknesses & What We'd Improve

- The explanatory power of this model is still limited: Since factors such as ticket prices, marketing, game times, and short-term events in cities have not been fully incorporated, more variables should be added and nonlinear or hierarchical structures should be considered to enhance the explanatory power.
- The sample of this model may be unbalanced: Some years or groups may lack representativeness. Future work should handle exceptional years more clearly and adopt stratified cross-validation to reduce bias.
- The main valuation indicator of this model leans towards revenue multiples: This will make the response of cost shocks to the value end less direct. Future work should add EBITDA multiples and simplify DCF calibration, and strengthen the sensitivity to interest rates and growth

assumptions.

- The risk characterization of this model is static: Factors such as injuries currently resemble exogenous constants. Future work should change this to a probability process and link it with rolling decision-making, allowing risks to continuously enter the optimization objective.

11.3 Conclusion

Under the current data and assumptions: maintaining a stable lineup + maintaining a cautious approach to leverage, it is possible to balance profit and value while meeting cash and leverage constraints. The framework can be applied to other teams/leagues (by re-estimating a few parameters to reuse). Multiple valuation comparisons also show that aggressive trading may "look good" in terms of valuation, but it will increase costs and debt, weaken cash flow security, and overall is not advantageous; management should prioritize stable growth under the condition of cash and risk control, focus on box office/rating risks, and use changes in cost discipline and financing environment as adjustment triggers.

XII. References

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A letter to Owners and General Manager of the Indiana Fever Team

TO: Owners and General Manager, Indiana Fever

FROM: Team 2608869

Date: Feb. 2, 2026

SUBJECT: Practical Advice: Balance Wins, Revenue, and Risk with Clear Trigger Rules

We are submitting you a brief suggestion letter, highlighting that the team should strike a balance between "winning" and "making money/controlling risks", and providing actionable rules for implementation.

Our core suggestion can be summed up in one sentence: Generally, focus on "stability" (do not overly increase actions), and stabilize the revenue through ticket sales; only make adjustments when risk signals are triggered, and prioritize protecting cash and controlling debt.

To be specific, use the game data to estimate the team's strength and winning trend; then, combine this information (along with the opponent's popularity, etc.) to calculate "the expected number of spectators"; next, use dynamic ticket prices to convert the spectators into box office revenue; the box office and other incomes go into the financial ledger, after deducting wages and operating costs, the profit and cash are obtained; then, use the valuation method to determine the team's value; finally, use the "ratio of liabilities to value" to measure the risk, and use a set of borrowing and repayment rules to bring this ratio back to the target, while not touching the cash safety line and debt limit.

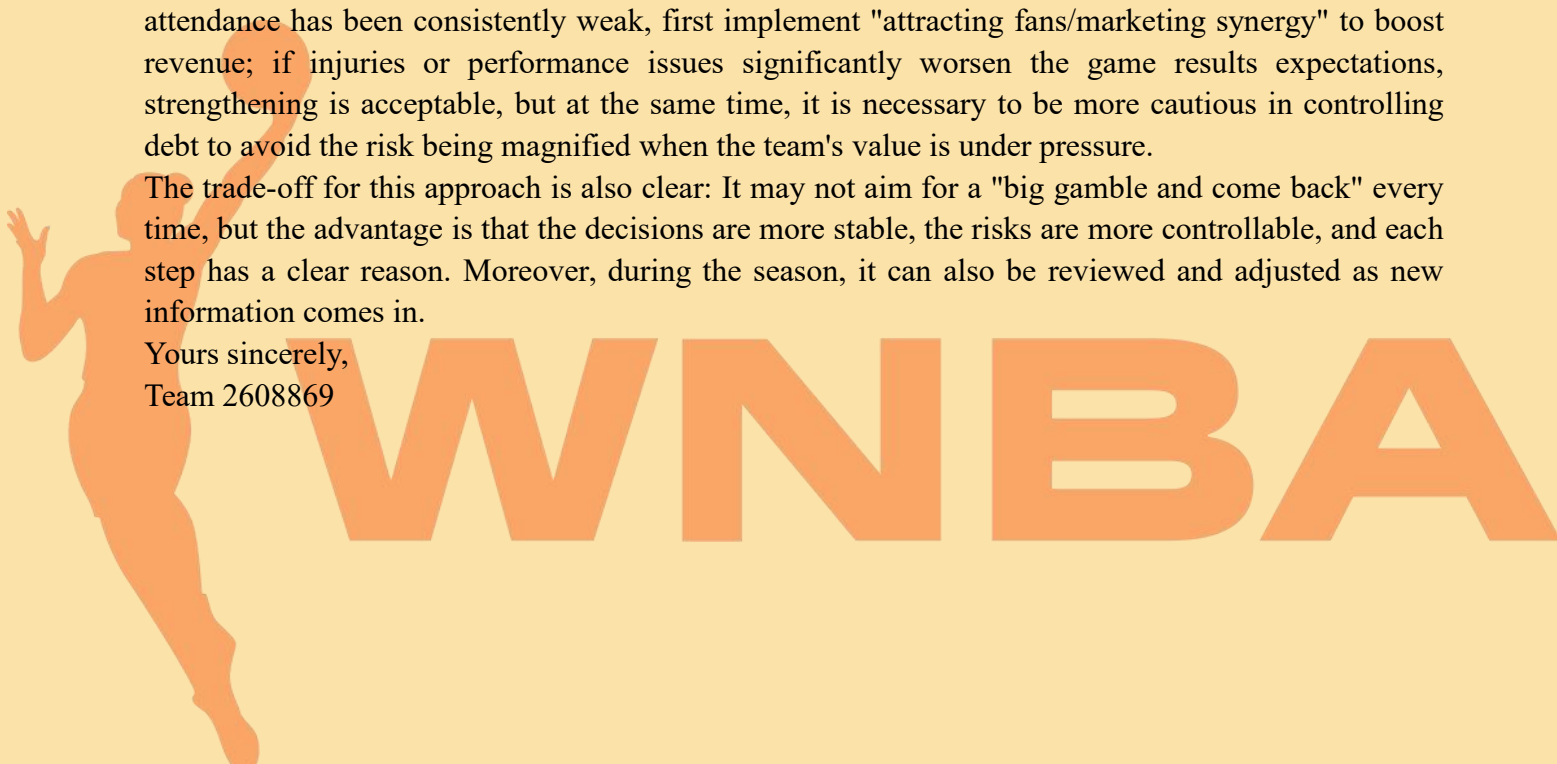
Based on this chain of reasoning, our advice is simple and easy to implement: Under normal circumstances, focus on "stability" and avoid frequent radical actions; in terms of operation, treat ticket sales as the most direct lever, implement "dynamic pricing" - for more crowded venues, moderately increase the ticket price, and for less popular venues, reduce the price and offer promotions to stabilize attendance and cash flow; financially, always maintain a sufficient cash safety cushion, keep the debt level moderate, and do not rely on leveraging to bet on short-term results.

When it is truly necessary to "take action", we recommend focusing on only three types of signals: If cash is tight, immediately tighten borrowing and spending, and set more conservative debt targets; if attendance has been consistently weak, first implement "attracting fans/marketing synergy" to boost revenue; if injuries or performance issues significantly worsen the game results expectations, strengthening is acceptable, but at the same time, it is necessary to be more cautious in controlling debt to avoid the risk being magnified when the team's value is under pressure.

The trade-off for this approach is also clear: It may not aim for a "big gamble and come back" every time, but the advantage is that the decisions are more stable, the risks are more controllable, and each step has a clear reason. Moreover, during the season, it can also be reviewed and adjusted as new information comes in.

Yours sincerely,

Team 2608869



AI Using Report:

1. Gemini 3 pro:

Quary; Explain in the simplest and most straightforward way what Monte Carlo simulation is.

Output: Monte Carlo simulation is a computerized technique used to predict the future when there is uncertainty involved.

In the simplest terms: **It acts like a "What If?" machine.**

Instead of just guessing one outcome, it runs a scenario thousands of times, changing the variables slightly each time, to show you all the possible outcomes and how likely they are to happen.

The Commute Analogy

Imagine you want to estimate how long it will take you to get to work tomorrow.

1. The "Single Guess" Method: You might say, "It usually takes 30 minutes." This is a single point estimate. It assumes everything goes normally. It doesn't account for rain, a flat tire, or hitting every green light.

2. The Monte Carlo Method: Instead of one guess, a computer simulates your drive **10,000 times**. In each simulation, it randomly tweaks the variables based on reality:

Simulation #1: It rains, traffic is heavy, but you hit green lights. (Result: 45 mins)

Simulation #2: It's sunny, no traffic, but you stop for gas. (Result: 28 mins)

Simulation #3: Major accident on the highway. (Result: 90 mins)

...and so on for 9,997 more times.

The Result

After running these thousands of "virtual drives," the simulation doesn't give you one number. It gives you a probability, like this:

10% chance you arrive in under 25 minutes.

80% chance you arrive within 30–40 minutes.

10% chance it takes longer than 50 minutes.

Why is it called "Monte Carlo"?

It is named after the famous casino in Monaco. Games of chance (like roulette or dice) rely on randomness. This method uses **random numbers** to simulate the uncertainty of real life, just like the randomness in a casino game.

Summary

Input: You define what *might* happen (ranges of possibilities).

Process: The computer plays out the scenario thousands of times.

Output: You get a map of risks and probabilities, rather than a single guess.

Would you like to see how a Monte Carlo simulation works for a specific scenario, like retirement planning or a project deadline?

2. Gemini 3 pro:

Quary: Please explain this question to me in the simplest and most understandable way. What is the question asking and what is my task?

Output: In simple terms, this problem asks you to act as a **business and strategy consultant** for a professional sports team owner.

The core conflict is that sports teams have two goals that don't always align: **winning games** and **making money**. Your job is to build a mathematical plan (a model) that helps the owner achieve both.

1. The Setup: Pick Your Team

First, you must choose a specific professional sports team to analyze.

It can be any sport, but it must be a team sport with at least **5 players playing at once** (like basketball, soccer, or football).

It must be part of a professional league.

Note: The problem mentions the WNBA as an example of a league facing big financial changes, but you are not required to use a WNBA team.

2. Your Main Task: Build a "Decision Machine"

You need to create a **dynamic model**. Think of this as a calculator or a flowchart that helps the owner make hard choices when things change.

The Goal: Maximize profit (money) *and* value (team prestige/worth) while managing the team's performance.

The Inputs: Your model needs to look at things like revenue (tickets, merchandise), expenses (salaries, taxes), and risks (losing streaks, economic downturns).

3. The Specific Problems You Must Solve

Once you have your model, you have to use it to answer these specific scenarios:

Recruitment Strategy: How should the team acquire players for next season? Should you use the draft, trades, or free agency?.

League Expansion: Imagine the league decides to add new teams. How does this change your strategy? Does a new team in a specific location hurt or help you?.

The "Wildcard" Decision: You must choose **one** additional business issue to model. Examples include:

Tickets: Should you sell cheap tickets to get full stadiums, or expensive tickets for higher profit?.

Venue: Should you renovate the stadium or build a new one?.

Media: How can you make more money from TV or streaming deals?.

Player Equity: Should you offer players a % ownership of the team instead of just cash?.

Injury Crisis: How does your model adjust if a key player gets injured? Does it tell you to panic, trade, or wait it out?.

4. What You Need to Submit

Your final submission (PDF) is a report of up to **25 pages**. It must include:

The Mathematical Solution: All your models, data, and strategies.

A Letter to the Owner: A 1-2 page letter written in plain English to the team owner. Summarize your strategy, the risks, and how your plan makes them money while keeping the team competitive.

In short: You are building a simulation that tells a team owner, *"Here is how we spend our money to win the most games and make the most profit, even if the league changes or players get hurt."*

3. Gemini 3 pro:

Quary: What does the term "leverage" mean in the question? Is it the "debt/borrowing" kind of leverage? How should I model it?

Output: Based on the problem statement, the term "leverage" most likely refers to Financial Leverage (the "debt/borrowing" kind), but it can also be interpreted as Strategic Leverage (bargaining power).

Here is the breakdown of why this is the case and how you should model it.

1. Is it the "Debt/Borrowing" kind? YES.

The text explicitly links the owner's decisions to financing and debt. This is the classic business definition of leverage: using borrowed money (debt) to increase the potential return of an investment (the team).

Evidence: The problem states that "Every season, the owner must decide how much to finance with debt versus equity".

Evidence: It also mentions that owners face risks that may cause them to "take on substantial debt".

The Connection: When the problem asks you to help the owner "adjust their leverage", it is asking you to determine how much debt the owner should take on to pay for better players or facilities.

2. The Secondary Meaning: Strategic Leverage

In sports business, leverage also means negotiating power. A winning team has more "leverage" to demand better deals.

Evidence: The text asks about establishing methodologies for "setting offers, negotiating, and writing contracts".

Evidence: It mentions "negotiations and demands" regarding revenue sharing.

The Connection: If your team is winning (high performance), you have "leverage" to charge more for tickets or demand better media deals.

How to Model "Leverage"

Since you need a "dynamic decision-making model", you should treat leverage as a control variable (a dial the owner can turn).

Option A: The Financial Model (Primary Focus)

Model leverage as the Debt-to-Equity Ratio.

The Decision: The owner decides how much money to borrow to buy expensive players (inputs).

The Trade-off:

High Leverage: You borrow heavily to sign stars. If they win, revenue spikes, and you pay off the debt easily (high profit).

The Risk: If the players get injured or underperform, you still owe the debt but don't have the revenue to pay it. Your model must calculate this risk.

Option B: The Strategic Model (Secondary Focus)

Model leverage as a Multiplier of Influence.

The Decision: Use your team's popularity/wins to "lever" higher prices.

The Formula: Negotiating Leverage = function(Win Rate, Star Power).

Application: If Win Rate is high, the "Leverage" variable increases, allowing you to set Ticket Price higher without losing attendance.

Recommendation: Focus on Financial Leverage (Debt) as your main mathematical variable, because the prompt explicitly discusses "financial modeling" and "debt versus equity". However, you can make your model sophisticated by showing how Strategic Leverage (winning games) helps you pay off that Financial Leverage (debt).

4. Gemini 3 pro:

Quary: Which sports/teams can I choose? Are there any strict requirements?

Output: You have a lot of freedom in choosing your team, but there are three strict requirements you must follow according to the problem statement.

The 3 Strict Requirements

Minimum Player Count: The sport must have at least 5 players who play cooperatively at the same time.

Professional Status: The team must be a member of a professional league. You cannot choose a college, amateur, or recreational team.

Data Availability: You must be able to find publicly available sport and finance data for the team you choose.

Which Sports Can You Choose?

As long as they meet the criteria above, you can choose almost any major team sport.

✔ Allowed:

Basketball (5 players)

Soccer / Football (11 players)

American Football (11 players)

Baseball (9 players)

Ice Hockey (6 players)

Volleyball (6 players)

Rugby, Cricket, Lacrosse, etc.

✗ NOT Allowed (Too few players):

Tennis (Singles or Doubles)

Beach Volleyball (2 players)

Golf (Individual)

Bowling or Boxing (Individual)

3x3 Basketball (Only 3 players)

Do I have to use a WNBA team?

No. The problem uses the WNBA as a detailed example of a league going through financial changes , but it explicitly states:

"You may use a WNBA team if you care to, but you are not required to do that".

Summary Advice

Choose a team where you can easily find financial numbers (salary caps, ticket prices, revenue). Major US leagues (NBA, NFL, MLB, NHL, WNBA) or major European soccer leagues (Premier League, La Liga) are usually the safest choices because their financial data is often reported in the news.

5. Gemini 3 pro:

Quary: How can "winning the game" and "making money" be considered simultaneously in this question? Which one should I set as the primary goal?

Output: Based on the problem statement, here is how you should handle the conflict between winning and making money.

1. Which one is the Primary Goal?

Making Money (Profit) is the primary goal.

The problem text is very explicit about this:

"In professional sports business, the primary goal is to make money for the owner and not necessarily win games".

"The player's job is to make money for the owner".

Your specific task is to design a model where the goal is to "maximize team profit and value".

While fans care about winning, you are modeling from the owner's perspective, where the team is an "entertainment profit-generating business".

2. How to Consider Them Simultaneously

You cannot just ignore winning, because winning is usually the engine that generates the money. The two are connected in a loop:

Spending (Investment): You spend money on better players (higher salaries) to improve performance.

Performance (Winning): Better players lead to more wins.

Revenue (Return): Winning generates more fan interest, which leads to higher revenue from tickets, merchandise, and media.

The Conflict (The Trade-off): If you spend too little, the team loses, fans stop coming, and revenue drops. If you spend too much chasing a perfect "undefeated" season, your salary costs might become so high that they eat up all your profits, even if you win the championship.

3. How to Model This Relationship

In your mathematical model, you should treat Profit as the "Objective Function" (the thing you want to maximize) and Winning as a "Driver" or "Constraint."

Option A: The "Means to an End" Approach

Goal: Maximize Profit.

How: Your model calculates the "Optimal Win Rate."

Example: The model might find that winning 60% of games generates the highest profit. Trying to win 90% of games costs too much in salary, and winning only 30% kills ticket sales.

This aligns with the text's prompt to value team dynamics "in terms of the profit for the team owners".

Option B: The "Constraint" Approach

Goal: Maximize Profit.

Constraint: "Win Rate must stay above X%."

This ensures the team stays respectable and competitive (maintaining the "value" of the franchise) without spending recklessly.

This fits the requirement to support "both competitive success and financial health".

Summary: Your model should tell the owner: "Spend exactly \$X on players. This will likely result in Y wins. This number of wins will generate the maximum possible profit (\$Z)."